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MASSES, RADII, AND MAGNETIC FIELDS OF PULSATING X-RAY SOURCES: IS THE "STANDARD" MODEL SELF-CONSISTENT?

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ABSTRACT

We present prototype calculations for Her X-1 and 4U 0115+63 which are designed to test the "standard" model of a pulsating X-ray source for self-consistency. We construct "semiempirical" mass-radius (M-R_s) relations for these two X-ray pulsars based on observational data (luminosities, pulse periods, spin-up rates, cyclotron spectral features) and their theoretical interpretation in terms of the MHD accretion torque theory of Ghosh and Lamb. These semiempirical M-R_s relations depend only on physical processes occurring *exterior* to an accreting neutron star. We compare them to theoretical M-R_s relations derived by integrating the hydrostatic equilibrium (Oppenheimer-Volkoff) equation for representative equations of state which might characterize the condensed matter in a neutron star's *interior*.

Our results support, in general terms, the "fast rotator," dipolar, accreting neutron star model for both Her X-1 and 4U 0115+63. However, our detailed but preliminary study indicates that quantitative consistency between our theoretical picture of the interior and exterior of a pulsating X-ray source may still be lacking. The apparent discrepancy is more pronounced for very stiff equations of state. We suggest plausible modifications to existing theoretical models which might remove the discrepancy.

Subject headings: dense matter — equation of state — pulsars — stars: interiors — stars: neutron — X-rays: binaries

I. INTRODUCTION

Ever since their discovery in the early 1970s, X-ray pulsars have promised to be remote space laboratories for Earth-bound astrophysicists. Theoretical interest in X-ray pulsars stems from the diversity of physical phenomena characteristic of them. Since most X-ray pulsars are believed to be accreting neutron stars in close binary star systems, their study may lead to new insights on the physics of condensed matter above nuclear densities $\rho \gtrsim \rho_{\rm nuc}$, $\rho_{\rm nuc} = 2.8 \times 10^{14}$ g cm^{-3} (see, e.g., Pandharipande, Pines, and Smith 1976; Baym and Pethick 1979; Lamb 1981). Neutron star structure calculations require, in addition to a detailed knowledge of condensed matter physics near nuclear densities, the full strong-field machinery of relativistic gravity. One might therefore hope to learn more about strong-field gravity from observations of X-ray pulsars (see Brecher 1978). The atmospheric science of accreting neutron stars offers the opportunity to study plasma physics and radiation transport at a combination of extremely high temperatures ($T \approx 10^7$–10^8 K), radiative fluxes ($F_\nu \approx 10^{24}$–10^{26} ergs s^{-1} cm^{-2}), and magnetic field strengths ($B_s \approx 10^{12}$ gauss) that very likely will never be reproduced in terrestrial laboratories (see Zel'dovich and Shakura 1969; Shapiro and Salpeter 1975; Basko and Sunyaev 1975; see also Wasserman and Salpeter 1980, and references therein). Farther from the neutron star surface, magnetospheric instabilities arising in strong magnetic and gravitational fields are important in channeling material to the stellar surface (see Arons and Lea 1976, 1980; Elsner and Lamb 1976). Outside the magnetosphere, the two or three-dimensional MHD accretion disk flows that power X-ray pulsars and govern secular variations in their pulse periods pose formidable theoretical problems of wide-ranging astrophysical importance (see Pringle 1981, and references therein).

After more than a decade of observations, some of these basic theoretical issues are now addressed by experimental data on X-ray pulsars. Masses have been estimated for six of the known X-ray pulsars (Bahcall 1978; Rappaport and Joss 1981), and the mass of the binary (radio) pulsar has been determined quite accurately (Taylor and Weisberg 1982). These determinations set lower bounds on the maximum mass of stable neutron stars, thereby constraining proposed

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equations of state above nuclear density. Cyclotron features have been detected in the spectra of two X-ray pulsars, Her X-1 and 4U 0115+63. In spite of ambiguities in their interpretation, these features indicate intense surface magnetic fields ($B_s \gtrsim 10^{12}$ gauss) for both X-ray pulsars (Trümper *et al.* 1978; Wheaton *et al.* 1979). Ongoing, accurate X-ray timing experiments have measured spin-up rates for nine of the X-ray pulsars (Rappaport and Joss 1981). These data may be compared with the predictions of detailed MHD models of accretion flows onto magnetic neutron stars (Ghosh and Lamb 1979).

At present, existing observations of X-ray pulsars provide substantial support for the following "standard" picture. The mass determinations require a moderately stiff equation of state of nuclear matter which allows neutron star masses as high as $M \gtrsim 1.4\ M_\odot$. The combined data are not inconsistent with the hypothesis that all neutron stars have mass $M \approx 1.4\ M_\odot$, as might be expected from stellar evolution theory (see Iben 1974; Joss and Rappaport 1976; Arnett 1979). Dipolar magnetic fields extend to the accretion disk, far from the surface of the X-ray pulsar. When extrapolated to the surface, these fields have strengths $B_s \approx 10^{11}$–10^{13} gauss, similar to the surface fields of radio pulsars. Steady accretion at the magnetospheric boundary causes X-ray pulsars to spin up.

In this paper we assess the self-consistency of this simple standard model. We do this for the two X-ray pulsars which exhibit cyclotron line features: Her X-1 and 4U 0115+63. For our prototype self-consistency check, we adopt the accretion torque theory proposed by Ghosh and Lamb (1979) as a concrete example. This theory successfully explains the combined spin-up data for the measured X-ray pulsars. In this theory the main spin-up torque on an X-ray pulsar is due to mass and angular momentum accretion from the inner edge of the disk at $r \equiv r_0$. In addition, there is either a spin-down or a spin-up torque due to the interaction of the pulsar's dipolar magnetic field with its accretion disk at $r > r_0$. In this theory, given the observed luminosity L, pulse period P_s, and spin-up rate $\dot{P}_s$ of an X-ray pulsar, one can deduce a relationship between its magnetic dipole moment μ, mass M, and radius R_s. If, as is assumed in the standard model, the neutron star magnetic field is purely dipolar, one can then obtain, entirely from the spin-up data, a prediction for the surface magnetic field B_s in terms of M and R_s.

The cyclotron line data may also be used to relate B_s, M, and R_s. Since the interpretation of the cyclotron line data is as yet controversial, we adopt the conservative point of view that the observed features in the spectra of Her X-1 and 4U 0115+63 are resonant *absorption* lines at 40 and 20 keV, respectively, formed at the magnetic poles. Upon equating the independent quantities

$$[B_s(M, R_s)]_{\substack{\text{spin-up}\\ \text{data}}} = [B_s(M, R_s)]_{\substack{\text{cyclotron}\\ \text{line data}}},$$

we get a mass-radius (M-R_s) relation for each of the two X-ray pulsars. We then compare this "semiempirical" M-R_s relation with the theoretical M-R_s relations derived by integration of the Oppenheimer-Volkoff (O-V) equation for selected nuclear matter equations of state. The intersection of the two M-R_s curves gives a unique mass and radius for the neutron star which we can evaluate in light of independent observational constraints on these parameters.

The two M-R_s curves for each star provide two very different physical perspectives from which to examine an X-ray pulsar. One curve is based on the *interior* behavior of a neutron star; in particular, it depends on the local equation of state of hadronic matter. The other curve is based on the *exterior* behavior of a neutron star; it depends on the nature of the plasma accretion flow onto the stellar surface and on the emission properties of plasma just above the surface. The shape of this "exterior" curve is determined by theory, but its normalization must be established by observations. In principle, the requirement that, when combined, the two perspectives yield a definite mass and radius for a given star which agree with independent observational determinations affords a powerful check on the self-consistency of our overall theoretical understanding of X-ray pulsars. In practice, however, the analysis reported here should be regarded as a "prototype calculation" at best, which is limited by the preliminary nature of some of the observational data (e.g., the cyclotron spectral features) and of the theoretical models (e.g., the equation of state calculations above $\rho_{\rm nuc}$, the restriction of the accreting magnetic neutron star calculation to aligned axisymmetric plasma flow, etc.). Hopefully the same analysis will be repeated and prove more useful in future years.

The main results of our prototype calculation are summarized in Figures 1 and 2. Our findings give strong qualitative support to the standard model in general and to Ghosh and Lamb's (1979) "fast rotator" models for Her X-1 and 4U 0115+63 in particular. However, satisfactory quantitative agreement between existing theory and observations can only be obtained for equations of state of nuclear matter which are not excessively stiff.

The basic theoretical ingredients of our analysis are presented in §§ II and III. To relate the magnetic dipole moment μ determined from accretion disk magnetohydrodynamics at $r \gtrsim r_0 \gg 2GM/c^2$ to a surface magnetic field at R_s ($\gtrsim$ a few times GM/c^2) we need the fully relativistic expression for a vacuum magnetic dipole field outside a spherical star. (Because the angular velocity of an X-ray pulsar is well below the break-up value, we can safely ignore nonspherical distortions due to rotation.) We derive general relativistic solutions for vacuum magnetic multipoles in an exterior Schwarzschild background in § II (see also Petterson 1975, and references therein). In § III we summarize the relevant

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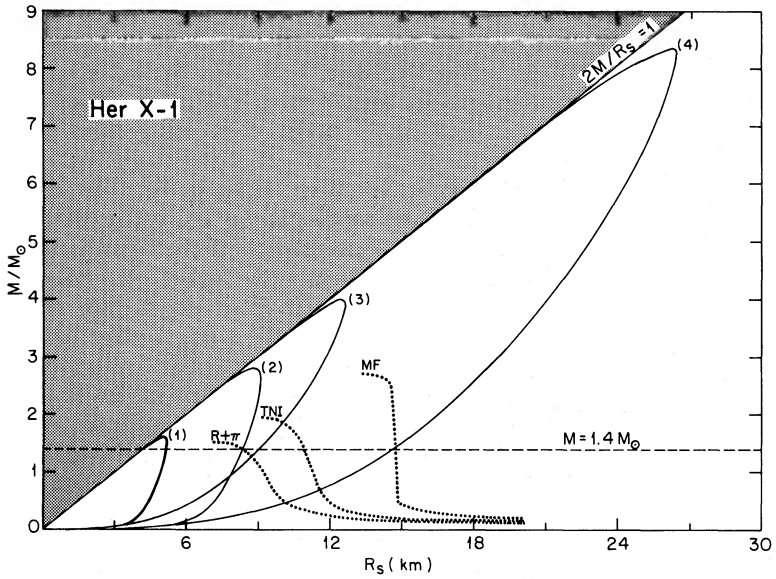


FIG. 1.—Comparison of interior and exterior mass-radius (M-R_s) relations for Her X-1. Numbered solid lines denote semiempirical, exterior M-R_s relations constructed as indicated in Table 1. Dotted lines are the theoretical interior M-R_s relations for the mean field (MF), three-nucleon interaction (TNI), and Reid + pion-condensate ($R + \pi$) equations of state. Dashed line is at $M = 1.4\ M_\odot$. Shaded area denotes the region of total gravitational collapse (i.e., black holes) from which static neutron stars are excluded.

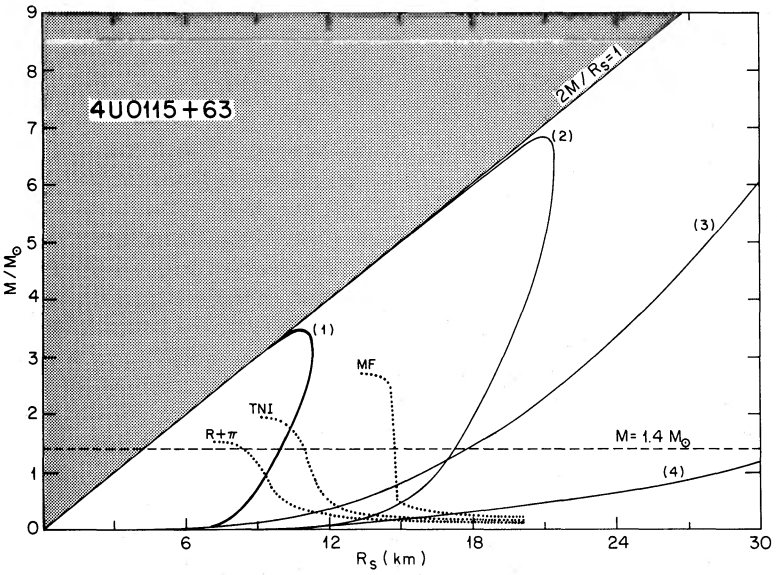


FIG. 2.—Comparison of interior and exterior mass-radius (M-R_s) relations for 4U 0115+63. All symbols have the same meaning as Fig. 1.

features of the accretion torque theory of Ghosh and Lamb (1979) so that we may incorporate the X-ray data on Her X-1 and 4U 0115+63 with the results of § II to obtain our M-R_s relation for each of these sources. We combine this curve with the M-R_s relation derived by integration of the O-V equation for an adopted equation of state in § IV. We construct four different "exterior" M-R_s curves corresponding to reasonable variations in the accretion torque parameters and three "interior" M-R_s curves corresponding to a soft, intermediate, and stiff nuclear matter equation of state. Finally, we apply our analysis in this section to assess the overall validity of the standard model for X-ray pulsars.

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II. MAGNETIC MULTIPOLE PROFILES IN AN EXTERIOR SCHWARZSCHILD BACKGROUND

Maxwell's equations in vacuum are

$$F^{\mu\nu}{}_{;\mu} = 0, \tag{1a}$$

where the field tensor $F_{\mu\nu}$ is related to the vector potential A_μ according to

$$F_{\mu\nu} = \frac{\partial A_\mu}{\partial x^\nu} - \frac{\partial A_\nu}{\partial x^\mu}, \tag{1b}$$

(see Misner, Thorne, and Wheeler 1972). We consider static, pure magnetic fields only, so that $A_0 = 0$ and $F_{\mu\nu}$ has spatial components only. We ignore any effects due to magnetospheric plasma. In addition, we consider azimuthally symmetric magnetic fields, for which we may write $\boldsymbol{A} = A^\phi(r, \theta)\boldsymbol{e}_\phi$. The only nonvanishing components of $F_{\mu\nu}$ are

$$F_{\phi r} = -F_{r\phi} = \frac{\partial A_\phi}{\partial r}, \qquad F_{\phi\theta} = -F_{\theta\phi} = \frac{\partial A_\phi}{\partial \theta}. \tag{2}$$

In a Schwarzschild background, the field equations (1), with $F_{\mu\nu}$ given in equation (2), becomes

$$r^2 \frac{\partial}{\partial r}\left[\left(1 - \frac{2M}{r}\right)\frac{\partial A_\phi}{\partial r}\right] + \sin\theta \frac{\partial}{\partial \theta}\left[\frac{1}{\sin\theta}\frac{\partial A_\phi}{\partial \theta}\right] = 0, \tag{3}$$

where we have adopted geometrized units with $G = c = 1$.

The separable solutions of equations (3) are

$$A^l_\phi = a^l_\phi(r)\left[-(1-\alpha^2)\frac{dP_l(\alpha)}{d\alpha}\right], \tag{4}$$

where $\alpha = \cos\theta$ and $P_l(\alpha)$ is the Legendre polynomial of order l. If we define

$$y \equiv 2M/r,$$

the radial equation for $a^l_\phi(r)$ becomes

$$\frac{d}{dy}\left[y^2(1-y)\frac{da^l_\phi}{dy}\right] - l(l+1)a^l_\phi = 0. \tag{5}$$

In the classical limit, $y \ll 1$, $a^l_\phi \approx y^l$, the familiar classical result for multipole fields (see eq. [13] below and Jackson 1972, eq. [5.41]). Let

$$a^l_\phi \equiv y^l \psi_l(y); \tag{6}$$

then from equation (5),

$$y(1-y)\frac{d^2\psi_l}{dy^2} + [2(l+1) - (2l+3)y]\frac{d\psi_l}{dy} - l(l+2)\psi_l = 0. \tag{7}$$

The solution that is regular at $y = 0$ is

$$\psi_l(y) = \frac{(2l+1)!}{(l-1)!(l+1)!}\sum_{n=0}^{\infty}\frac{(n+l-1)!(n+l+1)!}{n!(n+2l+1)!}y^n, \tag{8}$$

where we have taken $\psi_l(0) \equiv 1$.

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For $l = 1$ we can perform the series sum in equation (8), getting

$$\psi_1(y) = \frac{3}{y^3}\left[-\ln(1-y) - y - \frac{y^2}{2}\right]. \tag{9}$$

We therefore find for a dipole ($l = 1$) magnetic field

$$A_\phi(r,\theta) = \frac{3\mu \sin^2\theta}{2M}\left[x^2 \ln(1-x^{-1}) + x + \frac{1}{2}\right], \tag{10}$$

where $x \equiv r/2M$ and μ is a proportionality constant (recall eq. [3] is linear and homogeneous in A_ϕ). The nonvanishing components $F_{\mu\nu}$ can be found from equations (2) and (10). To relate $F_{\mu\nu}$ to the locally measured magnetic field, $\boldsymbol{B}$, we project it onto a local orthonormal tetrad, $e_{\hat{\lambda}}$. For the Schwarzschild metric this yields

$$F_{\hat{\phi}\hat{r}} = \frac{(1-x^{-1})^{1/2} F_{\phi r}}{r \sin\theta} = B^{\hat{\theta}} \tag{11a}$$

and

$$F_{\hat{\phi}\hat{\theta}} = \frac{F_{\phi\theta}}{r^2 \sin\theta} = -B^{\hat{r}}. \tag{11b}$$

Using equations (2), (10), (11a), and (11b), we find

$$B^{\hat{\theta}} = \frac{\mu \sin\theta}{r^3}\left[6x^3(1-x^{-1})^{1/2}\ln(1-x^{-1}) + \frac{6x^2(1-x^{-1}/2)}{(1-x^{-1})^{1/2}}\right], \tag{12a}$$

and

$$B^{\hat{r}} = \frac{2\mu \cos\theta}{r^3}\left[-3x^3 \ln(1-x^{-1}) - 3x^2\left(1+\frac{1}{2}x^{-1}\right)\right] \tag{12b}$$

(Similar results have been given by Petterson 1974). At very large r, $x = r/2M \gg 1$, and equations (12a) and (12b) become

$$B^{\hat{\theta}} \approx \frac{\mu \sin\theta}{r^3}, \tag{13a}$$

and

$$B^{\hat{r}} \approx \frac{2\mu \cos\theta}{r^3}, \tag{13b}$$

in agreement with the usual flat space results for a magnetic dipole if we identify μ as the magnetic moment.

The accretion torque on an X-ray pulsar is determined by physical processes occurring outside its magnetosphere, where $x \gg 1$ and equations (13a) and (13b) apply. Newtonian magnetohydrodynamics suffice in this regime, so we can use the Ghosh and Lamb (1979) theory, uncorrected for relativistic effects, to determine μ. However, to find the corresponding surface field strength at $x_s = R_s/2M \gtrsim 1$ we need the fully relativistic dipole formulae, equations (12a) and (12b). At the magnetic pole ($\theta = 0$), where the cyclotron spectral lines are presumably formed, the field is purely radial with a strength

$$B_s = \frac{3\mu}{4M^3}\left[-\ln(1-x_s^{-1}) - \frac{(1+x_s^{-1}/2)}{x_s}\right]. \tag{14}$$

Equation (14) will be used in § III to link the cyclotron line data, which probe the surface magnetic field strength and

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hence determine B_s, with the spin-up data, which probe the magnetic field strength far from the surface of an X-ray pulsar and hence determine μ.

III. THE SEMIEMPIRICAL "EXTERIOR" MASS-RADIUS RELATION

a) *Cyclotron Lines*

If B_s is the local value of the surface magnetic field, then the locally measured cyclotron transition energy in the stationary frame is

$$\Delta E_s = \frac{e\hbar B_s}{m_e c} f\left(\frac{B_s}{B_q}\right), \tag{15}$$

where

$$B_q = m_e^2 c^3/e\hbar \approx 4.4\times 10^{13} \text{ gauss}.$$

Generally, for either emission or absorption,

$$f\left(\frac{B_s}{B_q}\right) = 1 + \mathcal{O}\left(\frac{B_s}{B_q}\right). \tag{16}$$

For $B_s/B_q \lesssim 0.1$, we can set $f \approx 1$ to within 10% (see Wasserman and Salpeter 1980). The cyclotron transition energy corresponding to equation (15) measured by a distant stationary observer is

$$\Delta E_\infty = \left(1 - x_s^{-1}\right)^{1/2} \Delta E_s \approx \left(1 - x_s^{-1}\right)^{1/2} \frac{e\hbar B_s}{m_e c}. \tag{17}$$

Solving for B_s, we obtain

$$B_s \approx \frac{m_e c}{e\hbar} \frac{\Delta E_\infty}{\left(1 - x_s^{-1}\right)^{1/2}}. \tag{18}$$

Note that equation (18) gives B_s in terms of an observable quantity (ΔE_∞), fundamental constants, and the surface redshift parameter $x_s \equiv R_s/2M$.

Although it is likely that the cyclotron line features observed for Her X-1 and 4U 0115+63 are formed at the magnetic poles of these X-ray pulsars, it is not known for certain if these are emission or absorption lines. As we will see below (eq. [28]), a lower bound on ΔE_∞ gives an upper bound on M for a given x_s. A higher value of ΔE_∞ would move all of the curves in Figures 1 and 2 down and to the left, thereby diminishing the allowed phase space for self-consistent models of Her X-1 and 4U 0115+63. Since ΔE_∞ is smallest if the cyclotron features are resonant absorption lines, we adopt the conservative values $\Delta E_\infty \approx 40$ keV for Her X-1 and $\Delta E_\infty \approx 20$ keV for 4U 0115+63 (Trümper *et al.* 1978; Wheaton *et al.* 1979).

b) *Spin-up Rates*

Ghosh and Lamb have considered in detail the accretion torques on a rotating X-ray pulsar with a dipolar magnetic field (see Ghosh and Lamb 1979, and references to earlier work). The magnetic and spin axes were taken to be aligned in their calculation. The neutron star was assumed to be accreting from a thin Keplerian disk which corotates in the equatorial plane of the star, perpendicular to the magnetic (and spin) axis. In this theory, matter couples to the pulsar magnetic field in a thin boundary layer at the inner radius of the accretion disk

$$r_0 \approx 0.52 r_A^{(0)} \approx 0.52 \mu^{4/7} (2GM)^{-1/7} \dot{M}^{-2/7}, \tag{19}$$

where $r_A^{(0)}$ is the characteristic Alfvén radius for spherical accretion and $\dot{M}$ is the accretion rate. Since the angular momentum per unit mass at r_0 is $(GMr_0)^{1/2}$, the accreted material adds angular momentum to the X-ray pulsar at a

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rate

$$N_0 = \dot{M}(GMr_0)^{1/2}, \tag{20}$$

which causes the pulsar to spin up. In addition to this "spin-up" torque, there is a torque, which may cause either spin-up or spin-down, due to magnetic stresses that arise when stellar field lines are twisted by orbiting plasma in the accretion disk.

The *net* accretion torque in the Ghosh and Lamb model depends on the "fastness parameter"

$$\omega_s \equiv \frac{\Omega_s}{\Omega_K(r_0)} = \frac{2\pi}{P_s}\left(\frac{r_0^3}{GM}\right)^{1/2}, \tag{21}$$

where Ω_s is the angular velocity of the X-ray pulsar and $\Omega_K(r_0)$ is the Keplerian angular velocity of the disk at r_0. Clearly $\omega_s \leqslant 1$ if accretion is to occur at all. To a good approximation the net accretion torque is found to equal

$$N = N_0 n(\omega_s), \tag{22}$$

where N_0 is given in equation (20), and

$$n(\omega_s) \approx 1.4(1-2.86\omega_s)(1-\omega_s)^{-1} \tag{23}$$

depends only on ω_s. The exact values of the numerical constants in equation (23) depend on specific choices for certain dimensionless boundary layer parameters and on the viscosity parameter α of the accretion disk, but only insensitively. Note that for the specific numerical constants in equation (23), N vanishes at a critical fastness parameter $\omega_c \approx 0.35$. Note also that $N \approx N_0$ for $\omega_s \ll \omega_c$ and that $|N| \ll N_0$ for $|1-\omega_s/\omega_c| \ll 1$. The key features of the Ghosh-Lamb accretion torque theory tested in our self-consistency check are that (1) $r_0 \approx r_A^{(0)}$ (see eq. [19]) and (2) $N \ll N_0$ for $\omega_s \approx \omega_c \approx 1$. Other accretion torque theories (see Arons and Lea 1976, 1980; Scharlemann 1978) which do not share these characteristics may be similarly tested.

Timing measurements (Rappaport and Joss 1981) indicate that Her X-1 and 4U 0115+63 both spin up "anomalously slowly" vis-à-vis equation (20); i.e., although $N > 0$, $N \ll N_0$. Thus, ω_s is near ω_c for both sources and depends only very weakly on their masses. In the following we will ignore the slight M dependence of ω_s. In fact, the model of Ghosh and Lamb gives $\omega_s \approx 0.35$ for Her X-1 and $\omega_s \approx 0.30$ for 4U 0115+63, both very close to $\omega_c \approx 0.35$.

Given ω_s, we can solve equation (19) and (21) for μ to find

$$\mu \approx 0.44(\omega_s P_s)^{7/6}(GM)^{5/6}\dot{M}^{1/2}. \tag{24}$$

Now the steady-state luminosity L measured by a distant observer is related to the mass accretion rate $\dot{M}$ by

$$L = c^2\dot{M}\left[1-\left(1-x_s^{-1}\right)^{1/2}\right], \tag{25}$$

where c is the speed of light and the appropriate relativistic redshift factor has been included. Equation (25) assumes only that the accreting matter is nonrelativistic at large distances from the star and comes to rest at the stellar surface. Equations (24) and (25) give

$$\mu \approx 0.44\frac{(\omega_s P_s)^{7/6}(GM)^{5/6}L^{1/2}}{c\left[1-\left(1-x_s^{-1}\right)^{1/2}\right]^{1/2}}. \tag{26}$$

If we substitute equation (26) for μ into equation (14) for B_s, we find (restoring G and c)

$$B_s \approx 0.33\frac{(\omega_s P_s)^{7/6}L^{1/2}c^5}{(GM)^{13/6}}\left[\frac{-\ln\left(1-x_s^{-1}\right)-x_s^{-1}\left(1+x_s^{-1}/2\right)}{\left[1-\left(1-x_s^{-1}\right)^{1/2}\right]^{1/2}}\right]. \tag{27}$$

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c) M-R_s Relations

Equations (18) and (27) are independent determinations of the polar field strength B_s. For a consistent model these two determinations must agree. On equating equations (18) and (27), we find that

$$\frac{M}{M_\odot} \approx 21 \frac{[P_s(\mathrm{s})\omega_s]^{7/13} L_{37}^{3/13}}{[\Delta E_\infty(\mathrm{keV})]^{6/13}} [F(x_s)]^{6/13}, \tag{28}$$

where P_s (s) is the pulse period in seconds, $L_{37} = L/10^{37}$ ergs s^{-1}, ΔE_∞(keV) is the observed cyclotron line energy in keV, and

$$F(x_s) \equiv \frac{(1-x_s^{-1})^{1/2}[-\ln(1-x_s^{-1}) - x_s^{-1}(1+x_s^{-1}/2)]}{[1-(1-x_s^{-1})^{1/2}]^{1/2}}. \tag{29}$$

Equations (28) and (29) give an implicit M-R_s relation that is easily solved numerically once the observational parameters P_s, ω_s, L_{37}, and ΔE_∞ are known. For each value of x_s, M is determined from equation (28) and R_s is determined from

$$R_s = \frac{2GM}{c^2} x_s. \tag{30}$$

Note that from equations (28) and (30) both M and R_s scale with the observational parameters in the same way. From equations (28) and (30) it is clear that the largest possible values of M and R_s are obtained for the smallest possible choice of ΔE_∞, as was already mentioned in § II*a*.

Although, from equations (28) and (30), M and R_s do not depend sensitively on L, we will want to allow the possibility that the true total luminosity of an X-ray pulsar is somewhat greater than its detected luminosity in the 2–20 keV band. For example, appreciable thermal emission may exist in the soft X-ray regime below 2 keV. Alternatively, the emission from an X-ray pulsar may be strongly and multiply beamed, so that any given observer only sees a small fraction of the total flux from the source. A plausible theoretical and empirical upper bound on L is the Eddington luminosity for spherical accretion

$$L_\mathrm{E} = \frac{4\pi cGM}{\kappa_\mathrm{T}} \approx 12.6 \times 10^{37} \frac{M}{M_\odot} \text{ ergs s}^{-1}, \tag{31}$$

where k_T is the Thomson opacity (see, e.g., Margon and Ostriker 1973). The critical Eddington luminosity for accretion is determined far from the surface of the accreting X-ray pulsar, where $r \gg 2M$, so equation (31) is relativistically correct (see Shapiro and Teukolsky 1982). If we use equation (31) for L in equation (28), we find a second implicit mass-radius relation

$$\frac{M_\mathrm{E}}{M_\odot} = 112 \frac{[P_s(\mathrm{s})\omega_s]^{7/10}}{[\Delta E_\infty(\mathrm{keV})]^{3/5}} [F(x_s)]^{3/5}, \tag{32}$$

where $F(x_s)$ is given in equation (29) and the subscript "E" is a reminder that this is the M-R_s relation for $L = L_\mathrm{E}$. The corresponding equation for the radius is equation (30) with M replaced by M_E. Note that for $L < L_\mathrm{E}$, equation (28) always gives $M < M_\mathrm{E}$ for a given x_s at fixed values of P_s, ω_s, and ΔE_∞. The $M_\mathrm{E}(R_s)$ curve determined from equation (32) thus provides a plausible outer boundary to the allowed phase space for models of X-ray pulsars.

IV. THE THEORETICAL "INTERIOR" M-R_s RELATIONS AND THE CONSTRUCTION OF SELF-CONSISTENT MODELS

The results of our calculations are shown in Figures 1 and 2 for Her X-1 and 4U 0115+63, respectively. The solid lines are the semiempirical mass-radius relations derived from equations (28) and (32), with the observational parameters ω_s, P_s, L, and ΔE_∞ chosen according to Table 1. The thick solid line (curve 1) in each figure is the "standard model" mass-radius relation.

Also shown as dotted lines in Figures 1 and 2 are the M-R_s relations for stable neutron stars derived for three representative equations of state of nuclear matter. "MF" denotes the Pandharipande and Smith (1975) mean field

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TABLE 1

SEMIEMPIRICAL M-R_s RELATIONS[a]

Curve	Luminosity (L)	Fastness Parameter (ω_s)
1	standard[b]	standard[b]
2	standard	1
3	L_E	standard
4	L_E	1

[a] P_s and ΔE_∞ are fixed. For Her X-1, $P_s = 1.24$ s and $\Delta E_\infty = 40$ keV. For 4U 0115+63, $P_s = 3.6$ s and $\Delta E_\infty = 20$ keV.

[b] Her X-1: $L = 10^{37}$ ergs s^{-1}, $\omega_s = 0.35$. 4U 0115 +63: $L = 0.9 \times 10^{37}$ ergs s^{-1}, $\omega_s = 0.30$.

theory calculation of the equation of state. The MF equation of state is very stiff and so results in relatively large maximum masses and associated minimum radii. By contrast, "$R + \pi$" is a very soft equation of state based on the Reid potential for nucleon-nucleon interactions modified by charged-pion condensation (see Baym and Pethick 1979, and references therein). Maximum masses and minimum radii are relatively low for the $R + \pi$ equation of state. "TNI" denotes a recently constructed equation of state which includes a phenomenological three-nucleon interaction in addition to the usual two-body nuclear forces (Friedman and Pandharipande 1981). This equation of state has the virtue that its underlying nuclear interactions can account for laboratory data on the ground state energy, equilibrium density, and compressibility of nuclear matter. The maximum mass is 1.96 $M_\odot$, and the minimum radius is 9.2 km in neutron star models constructed with the TNI equation of state. This equation of state is intermediate in stiffness between the MF and $R + \pi$ equations of state.

The horizontal dashed lines in Figures 1 and 2 are at $M = 1.4\ M_\odot$ which, as has been argued, may be the unique mass of all neutron stars actually occurring in nature (Iben 1974; Joss and Rappaport 1976; Arnett 1979). Existing mass determinations for X-ray pulsars do not invalidate this hypothesis (Bahcall 1978; Rappaport and Joss 1981). Significantly, the mass of the binary radio pulsar is $M = 1.42 \pm 0.06\ M_\odot$ (Taylor and Weisberg 1982). On the other hand, masses estimated directly from the combined X-ray and optical light curves conservatively allow $0.4 \leqslant M/M_\odot \leqslant 2.2$ for Her X-1 (Bahcall 1978), and perhaps more realistically suggest $1.0 \leqslant M/M_\odot \leqslant 1.9$ (Rappaport and Joss 1981).

Examination of curve 1 in Figures 1 and 2 together with the M-R_s curves for representative equations of state first reveals that, in a very general sense, the "interior" and "exterior" curves tend to occupy the same regions in the M-R_s diagram. That this result was by no means guaranteed is clear from the proportionalities M, $R_s \propto \omega_s^{7/13}$ (see eqs. [28] and [30]). Accordingly, gross inconsistencies would occur if the accretion torque models had suggested that these sources were slow rotators with $\omega_s \ll 1$. Instead, self-consistency with condensed matter calculations demands $\omega_s \approx 1$, in qualitative agreement with the fast rotator models of Ghosh and Lamb (1979). Moreover, consistent standard models for 4U 0115+63 are allowed for both the $R + \pi$ and TNI equations of state, although in neither case is M precisely 1.4 $M_\odot$.

For Her X-1, the best-studied X-ray pulsar, the standard model is clearly not as successful. None of the representative equations of state cross curve 1. Curves 2–4 demonstrates the effects of varying the parameters of the standard accretion torque model in a plausible manner. At fixed luminosity, $L = 10^{37}$ ergs s^{-1}, allowed models for Her X-1 should lie in the region between curves 1 and 2 if $0.35 \leqslant \omega_s \leqslant 1$. The precise value of ω_s, which depends on certain dimensionless parameters in the accretion torque theory, is somewhat uncertain, although $\omega_s \leqslant 1$ is required for accretion to occur. Only the very soft $R + \pi$ equation of state yields neutron star models for Her X-1 consistent with accretion torque theory with the standard luminosity but with $0.35 \leqslant \omega_s \leqslant 1$. At fixed $\omega_s = 0.35$ but with higher total luminosity in the range 10^{37} ergs s$^{-1} \leqslant L \leqslant L_E$, allowed models for Her X-1 must lie in the region between curves 1 and 3 in Figure 1. Both the $R + \pi$ and TNI equations of state yield consistent models in this region, although the allowed models for the TNI equation of state have masses closer to 2 $M_\odot$ than to 1.4 $M_\odot$. Finally, models consistent with the general features of the accretion torque theory may be constructed for all three equations of state under the rather unrestrictive assumptions $\omega_s \leqslant 1$ and $L \leqslant L_E$. These relaxed constraints on the accretion torque parameters allow models lying in the entire region in M-R_s phase space enclosed by curve 4. Models that have $M \approx 1.4\ M_\odot$ and are consistent with the very stiff MF equation of state are just barely included in this region.

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One can envisage two plausible physical situations in which our derivation of the standard accretion torque M-R_s relation (eq. [28]) would be invalid. First, a principal assumption of the accretion torque theory is that the magnetic and spin axes of an accreting X-ray pulsar are aligned and that the accretion disk lies in the equatorial plane. These assumptions are extremely convenient for the purposes of calculation. However, since the sources are pulsating, axisymmetry is clearly justified only approximately, if at all (Ghosh and Lamb 1979). For an aligned rotator the magnetic moment is defined to be

$$\mu = \left[B(r) r^3 \right]_{r \to \infty}, \tag{33}$$

and is the actual magnetic moment of the pulsar (see eq. [13a] and remarks immediately following). For nonaligned rotators, $B(r)$ in equation (33) must presumably be replaced by some suitably defined angular average at radius r in the accretion disk. Since B is smallest at the magnetic equator, $\theta = \pi/2$, for fixed r, the resulting value of μ is always larger than the true magnetic moment. If the true magnetic moment is $\mu/(1+\beta)$, then from equations (18), (27), (28), and (30), we get

$$M, R_s \propto (1+\beta)^{-6/13}, \tag{34}$$

which drives the standard model to smaller masses and radii, thus worsening the inconsistency between the "interior" and "exterior" models for Her X-1.

Another possibly invalid assumption of the standard model is that the magnetic field of an accreting binary X-ray pulsar is purely dipolar. The spin-up data allow us to estimate the *dipole* component of the surface magnetic field *only* (see § II*b*). The cyclotron data probe the magnitude of the *total* surface field (§ II*a*). If $B_s^{(\mathrm{total})} = QB_s^{(\mathrm{dipole})}$, then from equations (18), (27), (28), and (30), we get

$$M, R_s \propto Q^{6/13}. \tag{35}$$

According to equation (35), large values of Q drive the standard model to larger masses and radii, thereby improving the chances for self-consistent accretion torque-equation of state models for Her X-1. However, the required nondipolar contribution to $B_s^{(\mathrm{total})}$ must be appreciable. For example, a standard accretion torque model for Her X-1 constructed from the TNI equation of state would require $Q \approx 10$ for a mass $M \approx 1.4\ M_\odot$. Note that in this case the dipolar component of the field still dominates for $r \gtrsim 10 R_s$ and hence throughout the accretion disk. One can consider the alternative possibility that dipolar fields are completely unimportant (Flowers and Ruderman 1977.) In this case the inner radius of the accretion disk r_0 may be appreciably inside the Alfvén $r_A^{(0)}$, leading to a smaller accreted angular momentum, N_0 (see eq. [20]). As a result, there is a tendency to larger M and R_s for such models when compared to the standard model we have studied in detail. This trend may be easily shown to be insufficient to produce quantitative self-consistency.

Given our rather liberal allowances for deviations from the standard model, we tentatively conclude that, if existing accretion torque theory is roughly reliable, quantitative consistency between theory and observation can only be achieved for X-ray pulsars if the equation of state of dense nuclear matter is not very stiff. The data for Her X-1 appear to rule out the stiff MF equation of state except under very special circumstances, viz. $L \approx L_E$ and $\omega_s \approx 1$, which seem unlikely. Very soft equations of state with maximum masses greater than $1.4\ M_\odot$ are not excluded by the data on Her X-1. We feel that a promising equation of state for condensed matter in neutron stars can be constructed by incorporating pion condensation in the TNI nuclear interaction model. Such a calculation has not been performed as yet. We note parenthetically that pion condensation has already been suggested as one possible cure for the discrepancy between "standard" theoretical cooling curves for young neutron stars and *Einstein* X-ray data for three SN remnants (Cas A, SN 1006, and Tycho; see Nomoto and Tsuruta 1981, and references therein).

More significant than our preliminary conclusions is the fact that theoretical models of X-ray pulsars which address entirely different physical processes and characteristics can be tested for self-consistency. X-ray astronomy is on the threshold of making truly significant statements about the physics of highly condensed nuclear matter. We hope that the prototype calculation presented here may prove useful in this regard.

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